Iterative Planning with MUGS Explanations: Exploring the Design Space

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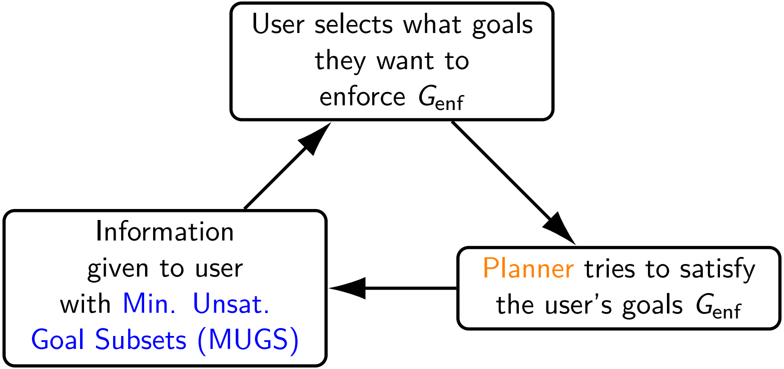
HAXP at ICAPS 2026

Iterative Planning Loop [Eifler et al., 2020a]



Research Questions:

1. do we really need a planner in the loop?
2. can we better visualise the MUGS?

Setting: Oversubscription Planning (OSP) following [Eifler et al., 2020a]

OSP task: $\tau = \langle V, A, c, I, G^{\text{hard}}, G^{\text{soft}}, b \rangle$

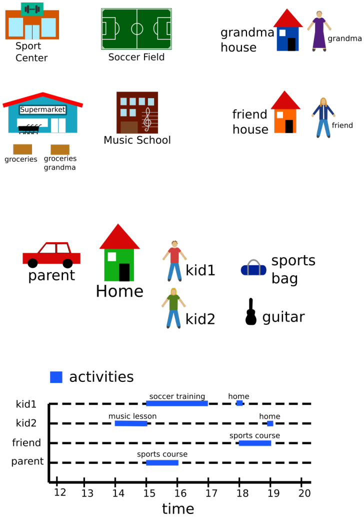
- variables V, actions A, and initial state I same as in a finite-domain representation (FDR) [Bäckström and Nebel, 1995; Helmert, 2009]
- action cost c
- hard and soft goals $G^{\text{hard}}, G^{\text{soft}}$
- b is an action-cost budget that disallows solutions whose cost exceeds b.

A solution for such tasks is a sequence of actions (called a plan) that can be applied to I, and lead to a state that satisfies G^{hard}. $I \xrightarrow{a_1} s_1 \xrightarrow{a_2} \ldots \xrightarrow{a_n} s_n$

We do not have utilities over G^{soft}!

Setting: Oversubscription Planning (OSP)

Example: Parent's Afternoon



Complete as many chores as possible

- **variables**: e.g., car-at location, in-car person
- **actions**: drive between locations with persons or objects in car
- **soft goals**: e.g., kid1 goes to soccer training, grandma gets her groceries

Steps of the Planning Loop

Step 1: User selects what goals they want to enforce

- Start with $\tau = \langle V, A, c, I, G^{\text{hard}}, G^{\text{soft}}, b \rangle$
- User selects $G_{\text{enf}} \subseteq G^{\text{soft}}$
- G_{enf} is enforced by setting hard goals $G^{\text{hard}} \cup G_{\text{enf}}$

Step 2: Planner tries to satisfy G_{enf}

2 possibilities:

1. does not find a plan
2. finds a plan that satisfies $G^{\text{hard}} \cup G_{\text{enf}}$

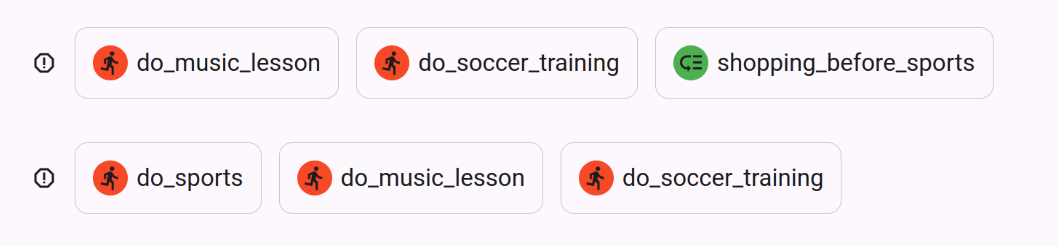
☞ useful information in both cases

Step 3: Information with MUGS

(Case 1: planner did not find a plan) User can ask *"why is there no plan?"* answer is a list of MUGS.

For $\tau = \langle V, A, c, I, G^{\text{hard}}, G^{\text{soft}}, b \rangle$ a subset $G \subseteq G^{\text{soft}}$ is a Minimally Unsolvable Goal Subset (MUGS) if the task with G enforced is unsolvable, but it is solvable for any $G' \subsetneq G$ [Eifler et al., 2020a].

Example: `do_music_lesson`, `do_soccer_training`, `shopping_before_sports`, `do_sports`



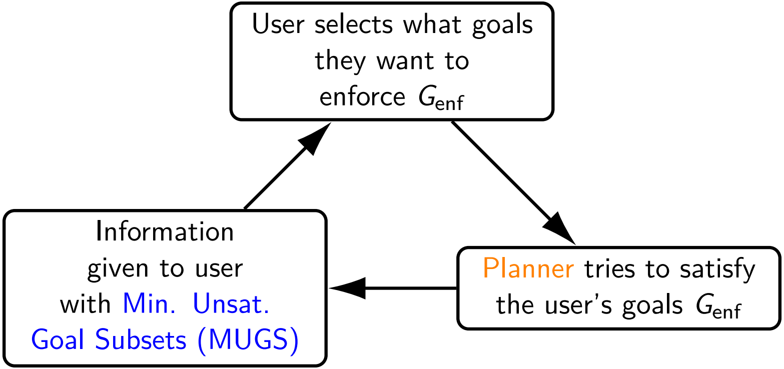
You must forgo at least one goal from each MUGS to make G satisfiable

Step 3: Information with MUGS

(Case 2: planner did find a plan π) user can ask *"why does π not satisfy p?"*

- if $G_{\text{enf}} \cup \{p\}$ can be satisfied: *"you can enforce this goal"*
- if $G_{\text{enf}} \cup \{p\}$ is unsatisfiable: a list of MUGS with the semantic *"you must remove from $G_{enf} \cup \{p\}$ at least one goal from each MUGS before $G_{enf} \cup \{p\}$ is satisfiable."*

Iterative Planning Loop (again)



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☞ do we really need a planner in the loop?

Currently: we run a planner to see if the user's selection G_{enf} can be solved

But the plan is not shown! Users only interact with *"why does π not satisfy p?"*

So we don't care about π, only if G_{enf} is satisfiable

Alternative: use MUGS to decide if G_{enf} is satisfiable!

$$G^{\text{hard}} \cup G_{\text{enf}} \text{ is satisfiable iff } \exists G \in \text{MUGS s.t. } G_{\text{enf}} \subsetneq G$$

(remember: G is MUGS if unsat but all strict subsets sat)

Important: we have MUGS "for free" because they are always precomputed a priori [Eifler et al., 2020a,b]

☞ do we really need a planner in the loop?

Tradeoffs

Using Planner

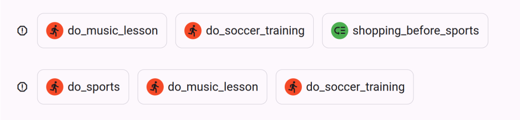
- produces an inspectable plan (but we don't inspect it)
- slow (mean 25 secs, up to 5 mins in our experiments)
- may satisfy additional goals

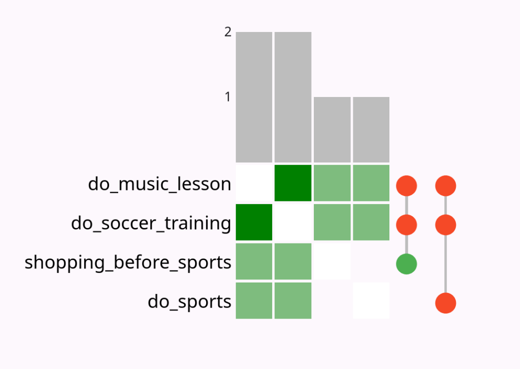
Using MUGS

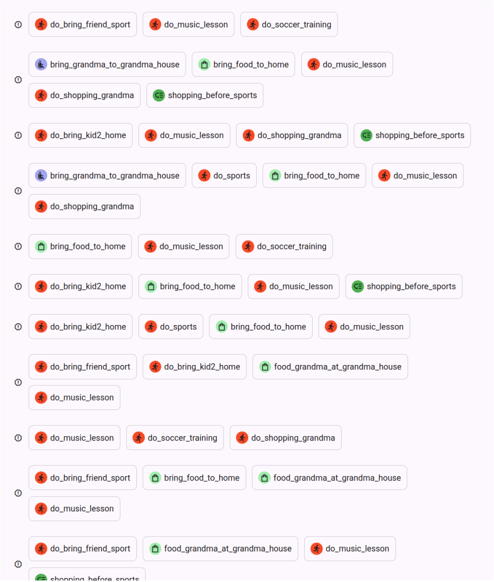
- does not produce a plan, only determines satisfiable or not
- fast (under a second)
- can not satisfy additional goals

Hypothesis: speedup of using MUGS outweighs the additionally satisfied goals

☞ can we better visualise the MUGS?






do_bring_friend_sport do_music_lesson do_soccer_training
bring_grandma_to_grandma_house bring_food_to_home do_music_lesson do_shopping_grandma shopping_before_sports
do_bring_kid2_home do_music_lesson do_shopping_grandma shopping_before_sports
bring_grandma_to_grandma_house do_sports bring_food_to_home do_music_lesson do_shopping_grandma
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do_bring_friend_sport do_bring_kid2_home food_grandma_at_grandma_house do_music_lesson
do_music_lesson do_soccer_training do_shopping_grandma
do_bring_friend_sport bring_food_to_home food_grandma_at_grandma_house do_music_lesson
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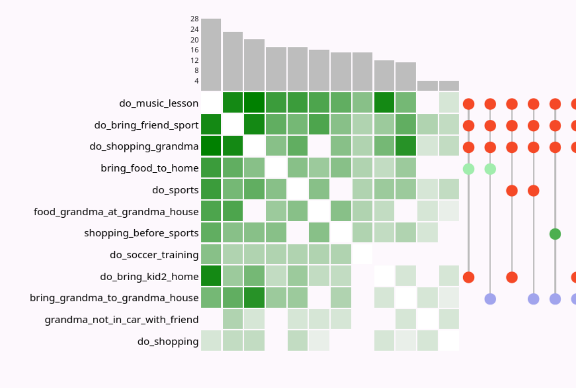

28
24
20
16
12
8
4
do_music_lesson
do_bring_friend_sport
do_shopping_grandma
bring_food_to_home
do_sports
food_grandma_at_grandma_house
shopping_before_sports
do_soccer_training
do_bring_kid2_home
bring_grandma_to_grandma_house
grandma_not_in_car_with_friend
do_shopping

User Study Design

Three groups

- no visualisation, with planner (`V-P+`)
- with visualisation, with plan (`V+P+`)
- no visualisation, no plan (`V-P-`) (4th option is future work)

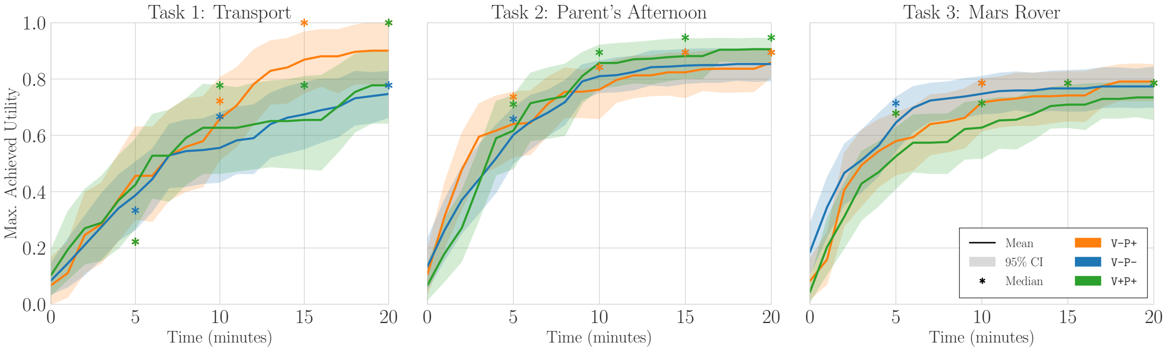
Three domains

- Transport (trucks deliver parcels, based on IPC NoMystery)
- Parent's Afternoon
- Mars Rover (based on IPC Rovers)

User Study Design

1. instructions and comprehension check
2. practice instance of Transport
3. evaluation instances with 20 minutes per instance, for Transport, Parent's Afternoon, and finally Mars Rover (in that order),
4. questionnaire

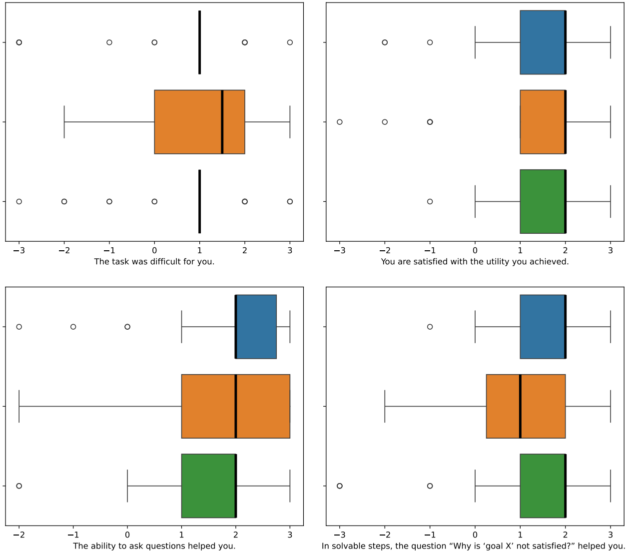
largely follows [Eifler and Hoffmann, 2022]



Only significant difference: Transport between V-P+ and V-P- from 12 minutes onwards, in favor of the baseline V-P+.

26–30 users per curve.

Likert Scale Responses



Only significant difference in favor of V+P+ over V−P+: (bottom right) *In solvable steps, the question “Why is ‘goal X’ not satisfied?” helped you*

Free Text Responses — *We didn't find any clear trends*

e.g., *"Please describe any limitations of the visualization in our tool and explain in what scenarios it was unhelpful."*

- "I think it's just visually distracting, takes some getting used to, it's complex"
- "It is very easy to navigate and work with. "
- "The table of unsolvable was hard to understand to find the solution"
- "EVERYTHING WAS HELPFUL ESPECIALLY THE WHY IS THIS UNSOLVABLE PART"
- "the fact that they is no chat bot to help"

Summary

Context

- oversubscription planning — a user must decide which subset of goals is most important
- MUGS explanations — "you must forgo goals X in order to make Y achievable"
- Iterative planning loop — user picks goals, planner tries to satisfy them, user receives MUGS

This work

- Do we really need a planner in the loop? *Probably not.*
- Can we visualise MUGS more effectively? *Probably yes, but. . .*

For more: `schmalz.cc/haxp-mugs`

Bäckström, C.; and Nebel, B. 1995. Complexity Results for SAS$^+$ Planning. *Computational Intelligence*, 11(4): 625–655.

Eifler, R.; Cashmore, M.; Hoffmann, J.; Magazzeni, D.; and Steinmetz, M. 2020a. A New Approach to Plan-Space Explanation: Analyzing Plan-Property Dependencies in Oversubscription Planning. In *AAAI*.

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